## INTRODUCTION

This article outlines instructions for the use of a new field alignment tool currently in use on MS7001FA and MS9001FA (single and two shaft) installations. Use of this tool is also recommended on future MS9001H and MS7001H single-shaft gas turbine installations.

These guidelines apply to a typical centerline installation, and do not supersede any factory instructions.

## HEALTH AND SAFETY REQUIREMENTS

In all cases, the Health, Safety and Environmental Manual (HSEM) will be the document followed on the job site during installation of this equipment.

Safety, health and environmental protection must be an integral part of all department business operations, including design, construction and start-up. Every reasonable effort will be made to create and maintain a safe and healthful work environment.

## REFERENCE DOCUMENTS

| No. | Description |
|---|---|
| MLI 0410 | Instructions, Alignment – Factory |
| MLI 0411 | Instructions, Alignment – Field Basic |

## DESIGN INFORMATION

The traditional field alignment method for gas turbines, load gears and generators has been to attach as set 0 to 1.000 inch dial indicators to take rim and face measurement readings on the coupling hubs while turning one of the rotors.

Although this method is practical for aligning smaller turbines, the larger FA and H Class units require the alignment to be performed earlier in the installation sequence (before assembling the lube oil and lift oil piping).

The method described in this article has been field proven and is recommended for alignment of MS7001FA and MS9001FA gas turbines. Because these units have 50 and 60 ton rotors, a new method is required to align the packages without having to turn the rotors. The concept of using a straight-edge alignment bar was derived from similar methods used by millwrights in the field for many years.

The alignment bar tool described in this article was designed, fabricated and first used in the 1993-1994 installation of four MS7001FA Gas Turbines at the Sithe Independence Power Station in Oswego, New York.

The installation schedule originally allotted 10 days to complete the generator alignment using dial indicators. Using the alignment bar method, the task was completed in only five days.

## REQUIRED TOOLS

### Alignment Bar Fixture

Figure 1 shows the measurements needed to fabricate an alignment bar. GE currently does not furnish this tool, but it can easily be made in the field by a machine shop.

### Other Required Tools

- One standard 6-inch “C” clamp.
- One Micrometer Depth Gage with 0 to 3.000 inch travel (Starrett Series No. 440 and No. 445).
- One Outside Micrometer Calipers with 0 to 1.000 inch travel (Starrett No. 436RL-1 inch).
- One Set of Adjustable Parallels – 6 parallels per set (Starrett No. S154LZ).

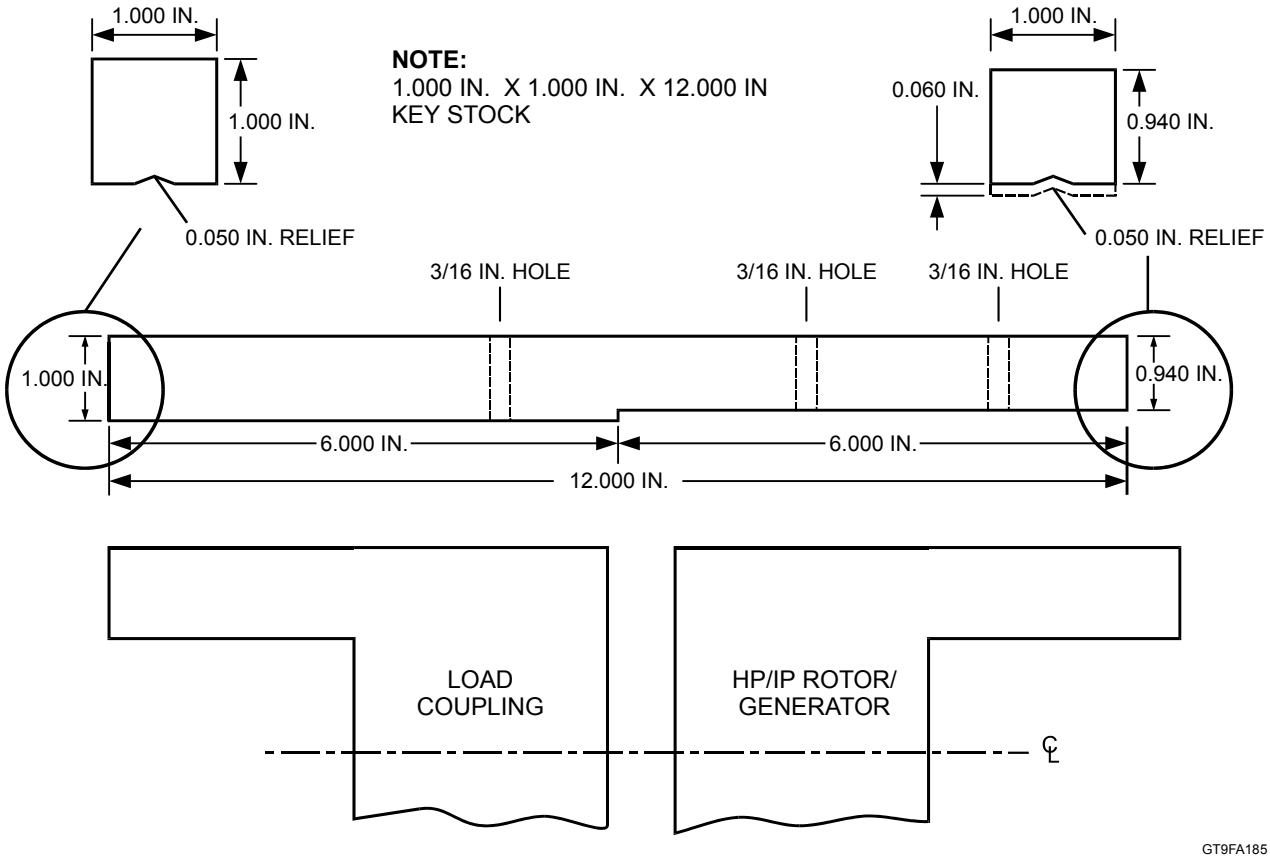



**Figure 1. Gas Turbine Coupling Alignment Tool – Typical Layout**

## ALIGNMENT METHOD

This alignment procedure must be performed at a point in the installation **before** the lube oil piping, lift oil piping, turning gear, and wiring have been installed.

**For MS7001FA, MS7001H, MS9001FA and MS9001H Single-Shaft Installations:**

1. Move the gas turbine into fine rough alignment using a straight-edge across the load coupling and HP/IP rotor hub. Proceed to Step 2.

**For MS7001FA and MS9001FA Simple Cycle or Two-Shaft Installations:**

1. Move the generators into fine alignment using a straight-edge across the load coupling and generator hub. Proceed to Step 2.

2. Use a "C" clamp to attach the alignment bar fixture (Figure 1) to the turbine load coupling flange OD at the 12, 3, 6 and 9 o'clock positions. Take a set of rim readings at each position (see Figure 2).

3. Use a depth gauge (Figure 3) to make a tightness check of the bar fixture to the coupling flange through a hole drilled in the fixture. (Subtract the thickness of the fixture from the measurement.)

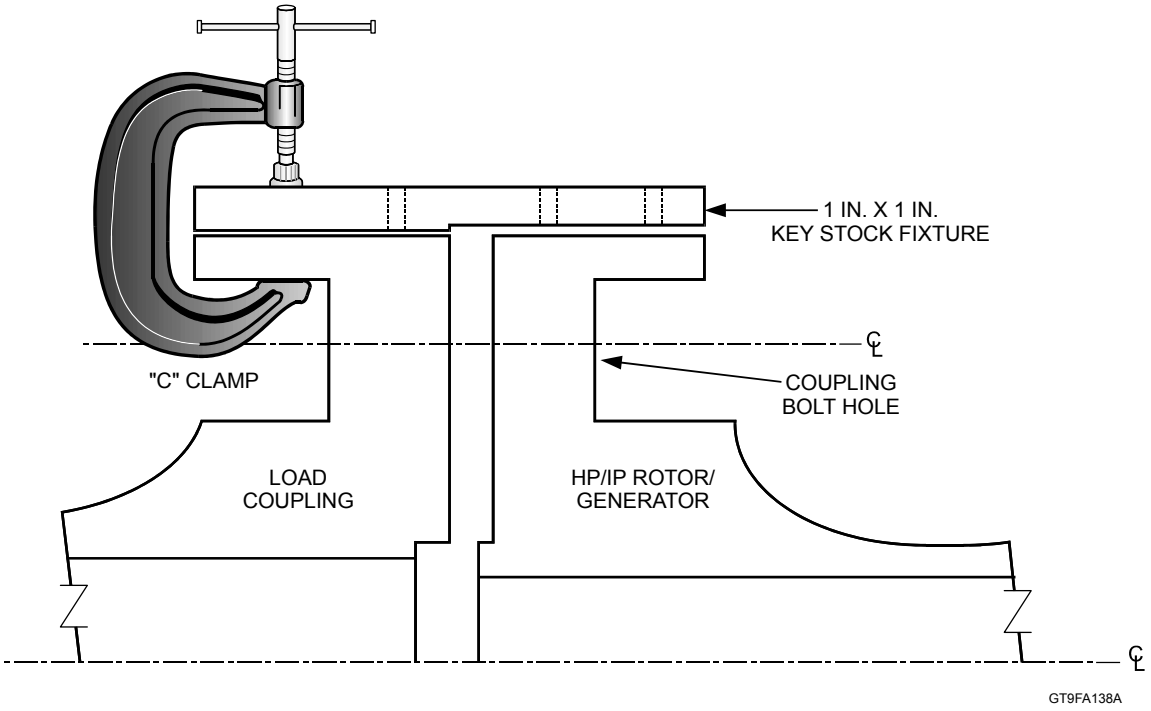
4. Move the depth gauge to the HP/IP rotor hub or generator end of the bar fixture and take the relative measurement through a hole drilled in the fixture (Figure 4). (Subtract the thickness of the fixture from the measurement for the rim reading at each position.)

5. Take face readings at the 12, 3, 6 and 9 o'clock positions using Adjustable Sliding Parallels and a 0 to1.000 inch Micrometer (see Figure 5).

**Note:** When final alignment has been reached on single-shaft turbine installations, the load coupling-HP/IP steam turbine spacer piece will be assembled. Refer to MLI 0411 for the procedure.

## TOOL VARIATIONS

A variation of the alignment bar has been made and used during installation of a MS7001FA Gas Turbine at FP & L and Tampa Electric with the same measurement accuracy. See Figure 6 and Figure 7.



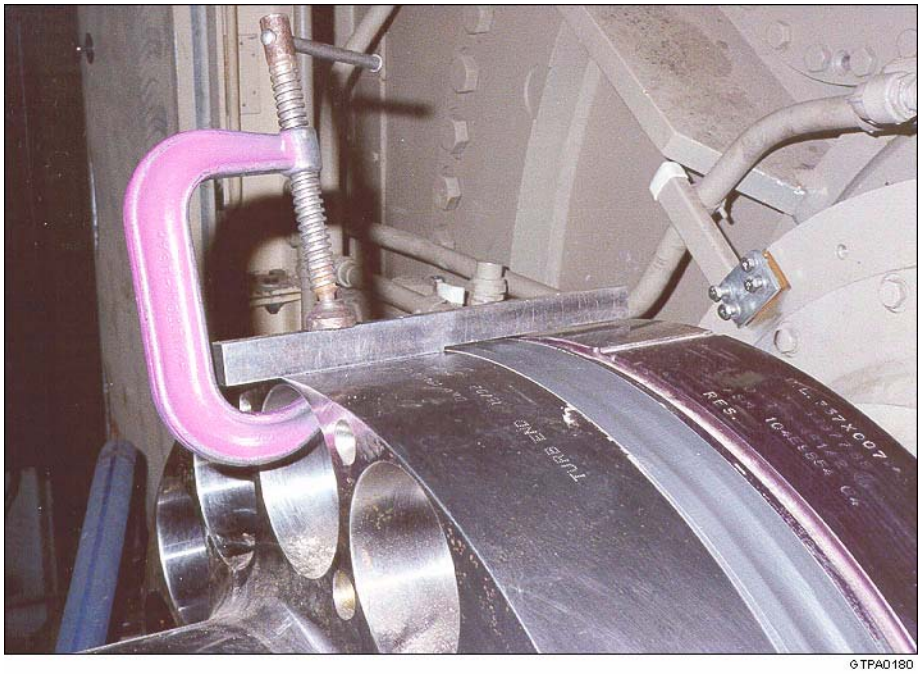




**Figure 2. Gas Turbine Coupling Alignment Tool – Typical Layout**

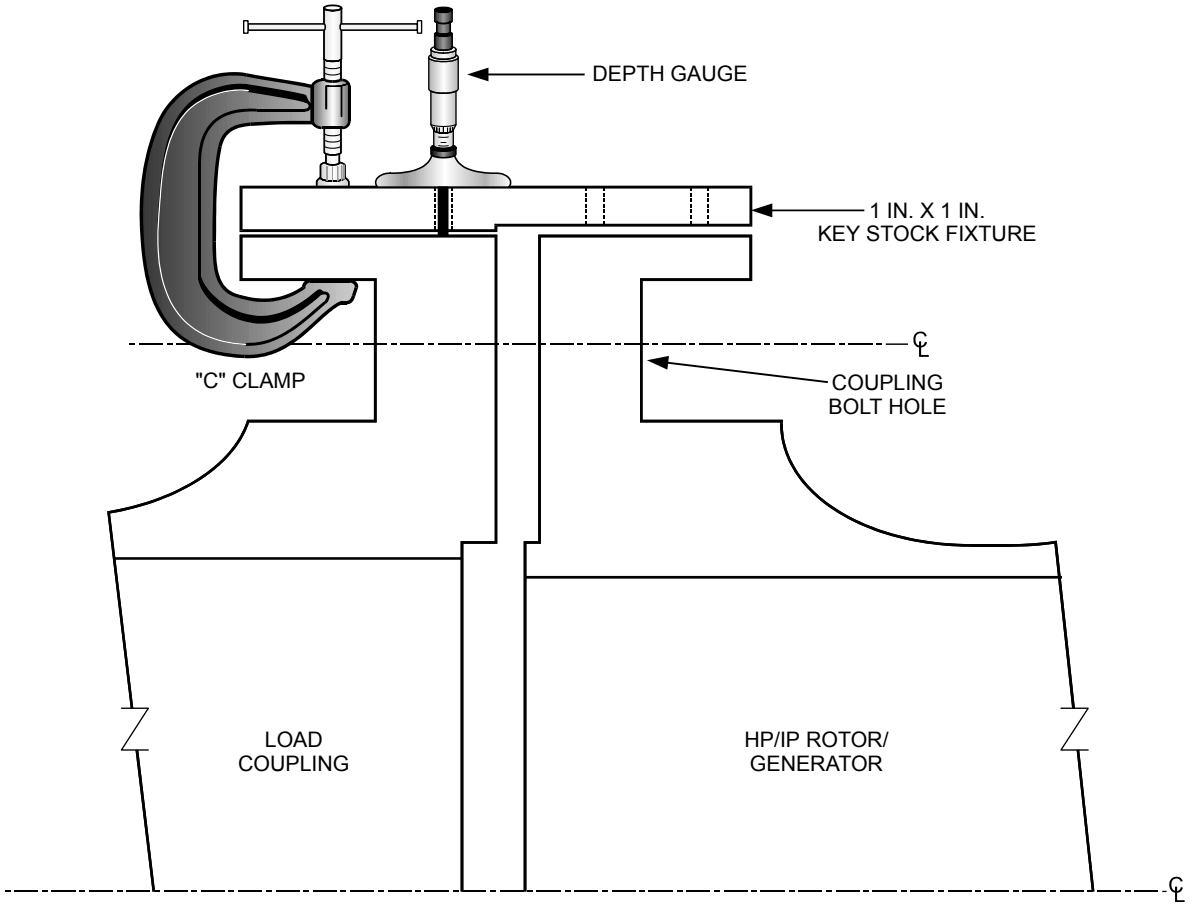
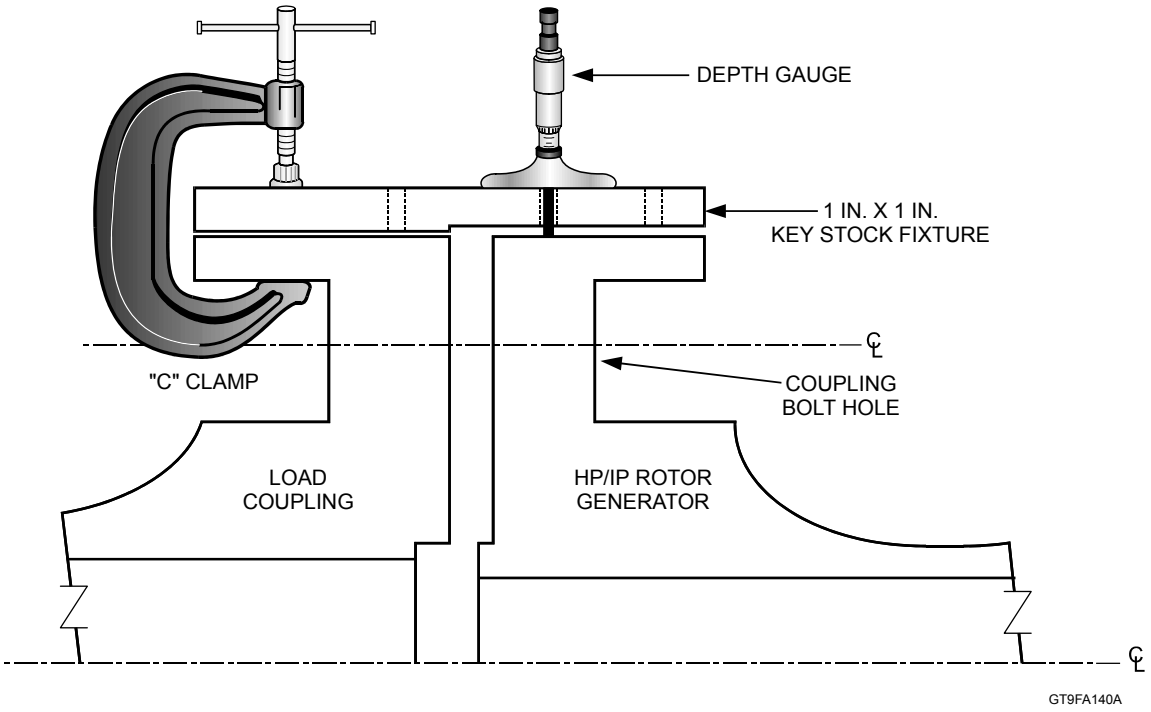



**Figure 3. Gas Turbine Coupling Alignment Tool – Typical Layout**



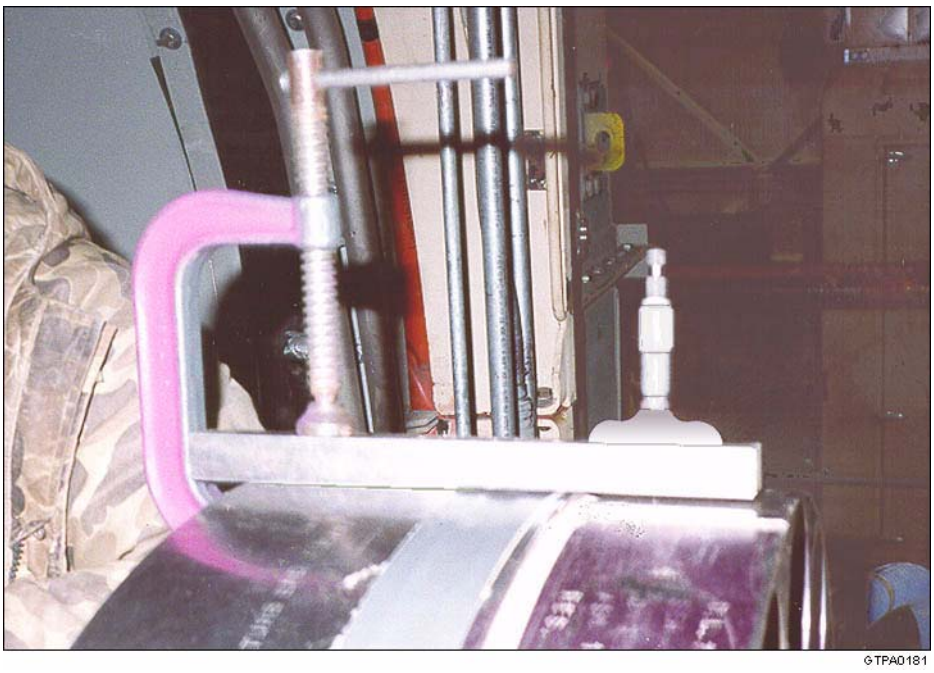




**Figure 4. Gas Turbine Coupling Alignment Tool – Typical Layout**

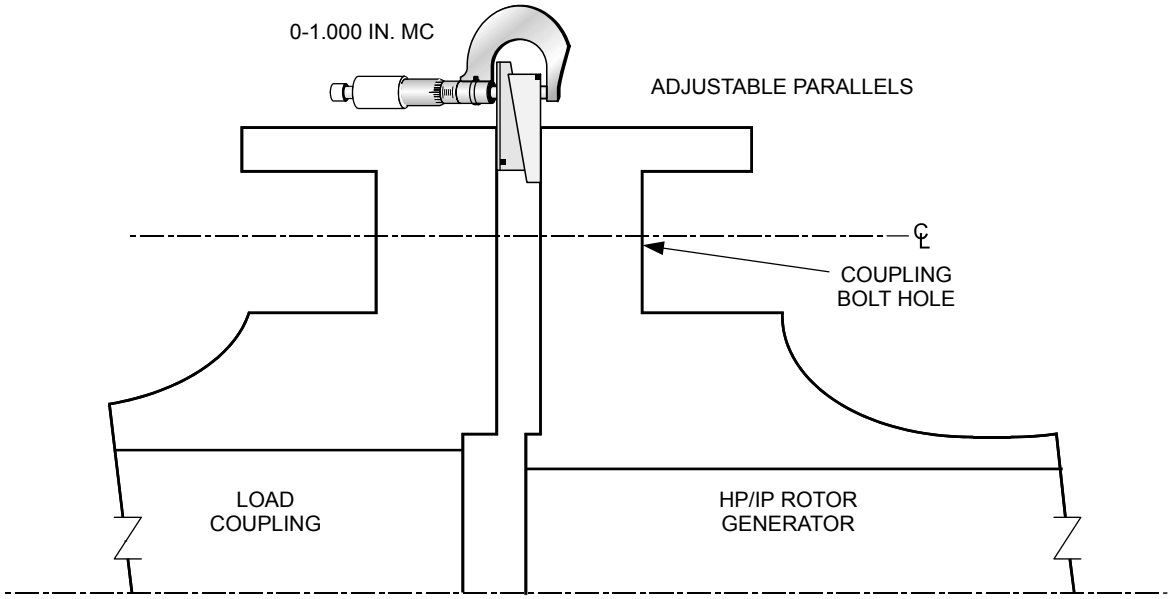






**Figure 5. Gas Turbine Coupling Face Alignment – Typical Layout**

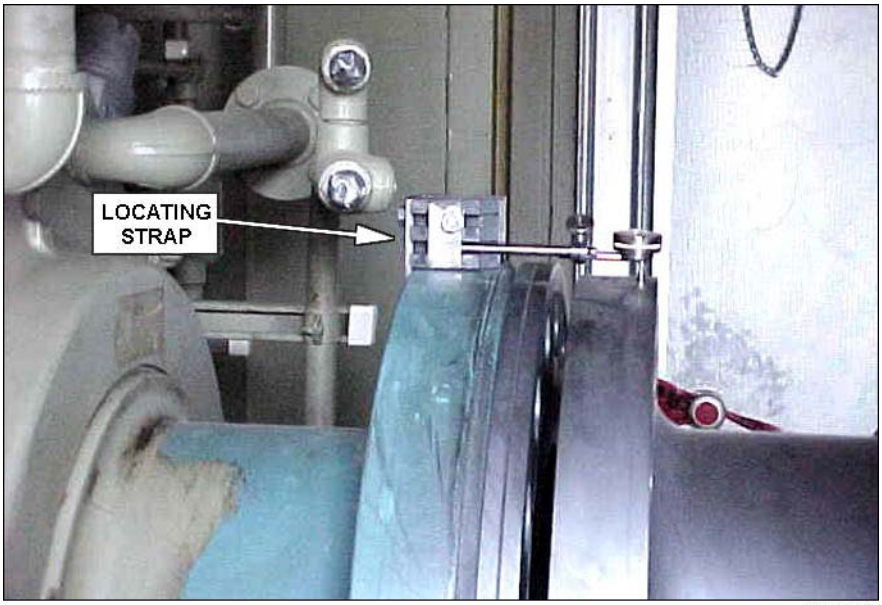



**Figure 6. Steel Block Shown Riding on Generator Hub with Local Dial Indicator Reading Load Coupling at 12 O'Clock Position**

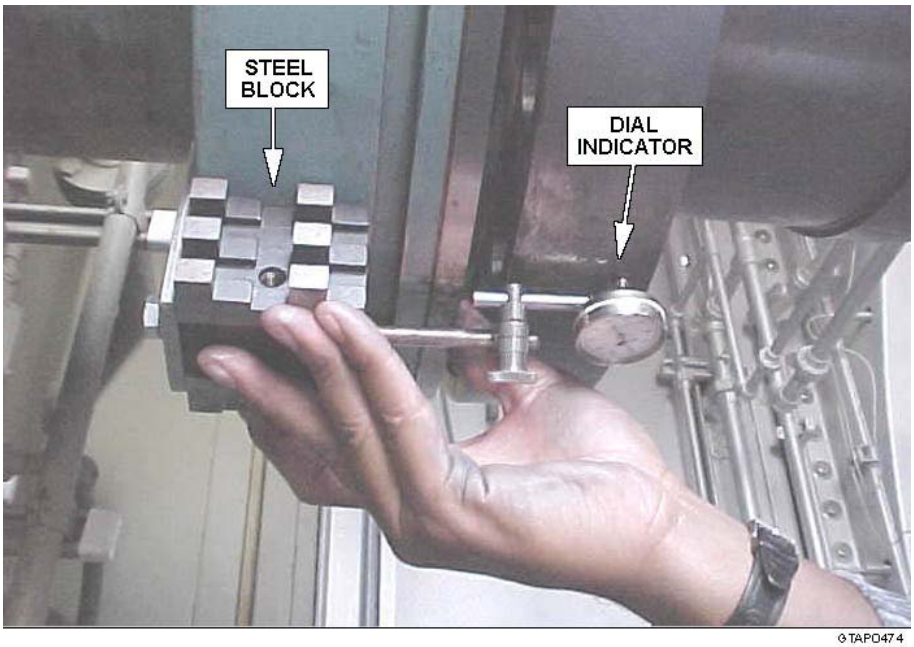



**Figure 7. Tool Shown at 6 O'Clock Position**